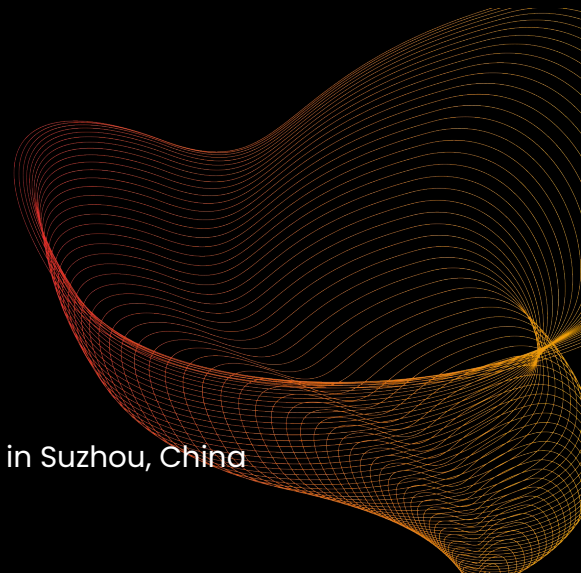


jina-embeddings-v4: Universal Embeddings for Multimodal Multilingual Retrieval

Saba Sturua, Elastic / Jina AI

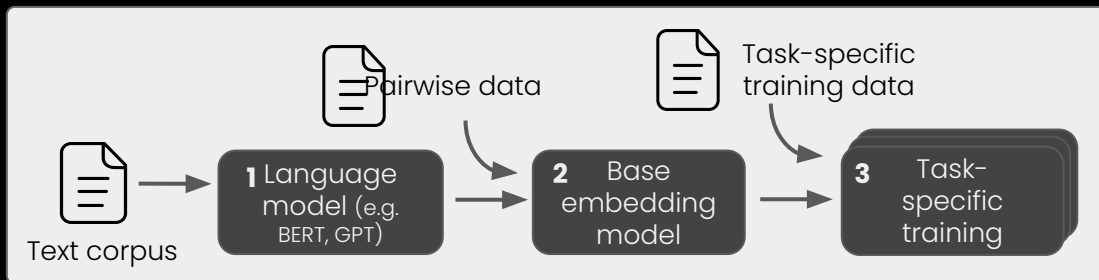
BoF "Embeddings, Rerankers, and Small LMs for Better Search" 7 Nov 2025 @ EMNLP in Suzhou, China



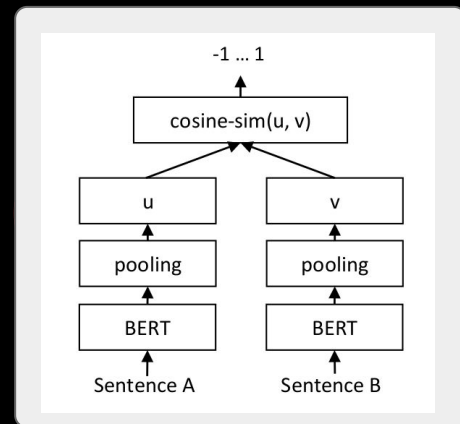
How are Vector Embeddings Trained?

Training Data Pipeline

- 1. Pre-training:** Masked language modeling on multilingual corpora
- 2. Fine-tuning:** Semantically-related text pairs with InfoNCE loss
- 3. Task-specific fine-tuning:** Classification, text matching, clustering, retrieval



Architecture



Source:
<https://www.sbert.net/docs/training/overview.html> (25.10.2023)

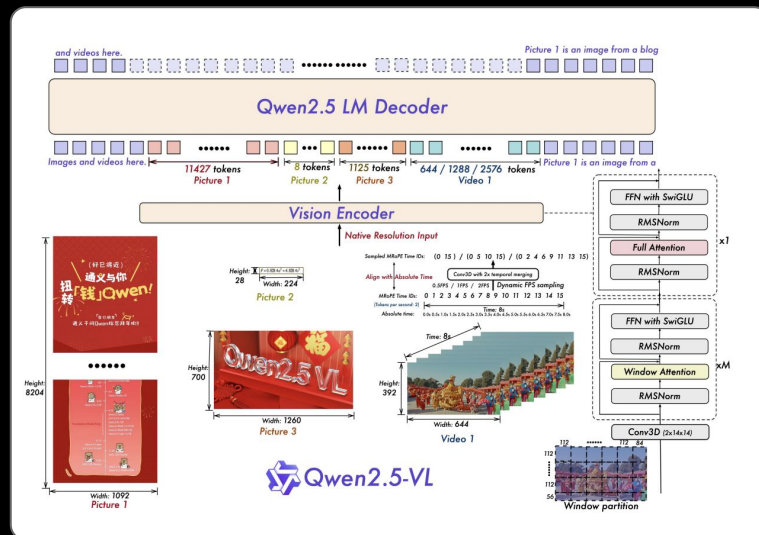
VLM Models: Architecture and Capabilities

Architecture

Combination of LLM + vision encoder (e.g. CLIP)

Capabilities

- Multimodal
- Multilingual
- Flexible resolution
- Trained on diverse domains, image types (pictures, illustrations, document screenshots, ...), and tasks (question answering, OCR, ...)



Source: <https://arxiv.org/pdf/2502.13923>

Jina Embeddings V4

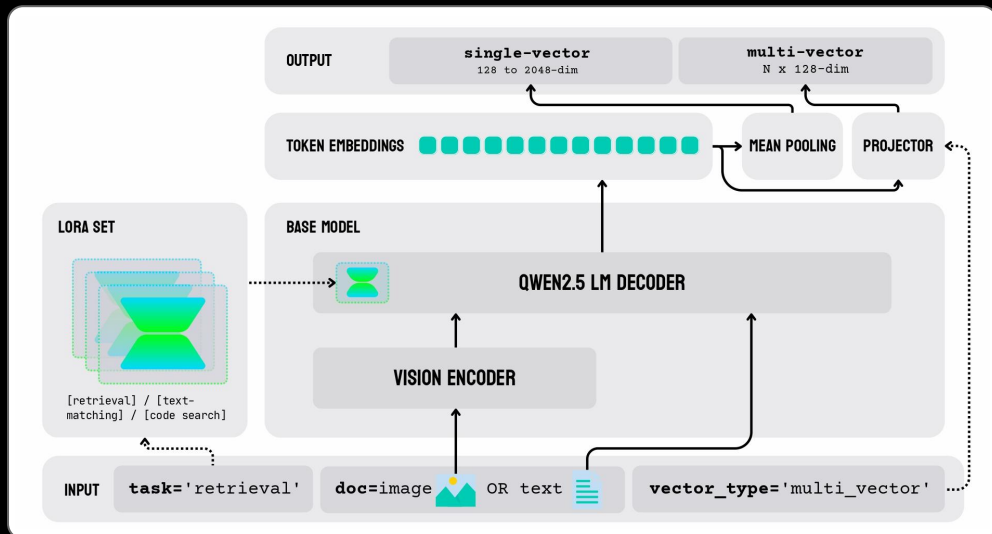
Multimodal: Text and image input

Multi-task: Retrieve text, documents, code + text similarity matching

Multilingual: 29+ languages

Optimized matching: Efficient single- and multi-vector retrieval

Flexible embeddings: Adjustable embedding dimensions

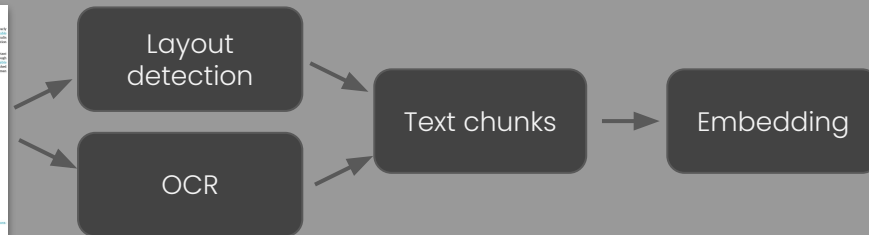


Visual Document Retrieval

Objectives

- **Encode documents as images**
 - Traditional OCR pipelines lose information
- **Dense (single-vector) or multi-vector search** (ColPali / CoBERT)

Traditional workflow:



Jina Embeddings V4 workflow:



Training Data

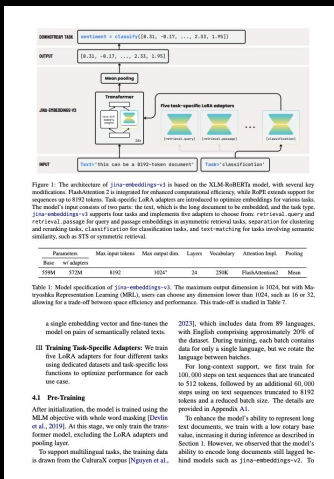
Text

"What historic landmark in Berlin symbolizes the reunification of East and West Germany?"

"The Brandenburg Gate is the historic landmark in Berlin that symbolizes the reunification of East and West Germany. Once a symbol of division during the Cold War, it now stands as a powerful icon of peace and unity in the heart of the city."

Document Images

"Show me jina-embeddings-v3 architecture"



Pictures

"Red sports car on a city street"



Training Loss

Contrastive training with multiple objectives

Retrieval methods

Dense Retrieval

$$s_{\text{dense}}(q, p) = \frac{\mathbf{q}_{\text{mean}} \cdot \mathbf{p}_{\text{mean}}^T}{\|\mathbf{q}_{\text{mean}}\| \|\mathbf{p}_{\text{mean}}\|}$$

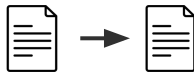
Late Interaction (Colbert)

$$s_{\text{late}}(q, p) = \sum_{i=1}^n \max_{j \in \{1, \dots, m\}} \mathbf{q}_i \cdot \mathbf{p}_j^T$$

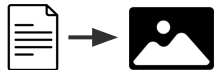
X

Modalities

Text to Text



Text to Image



→

General fine-tuning

Joint Loss Function

$$\begin{aligned} \mathcal{L}_{\text{joint}}(\mathcal{B}_{\text{txt}}, \mathcal{B}_{\text{multi}}, \tau) := & \\ & w_1 \mathcal{L}_{\text{NCE}}(\mathbf{S}_{\text{dense}}(\mathcal{B}_{\text{txt}}), \tau) \\ & + w_2 \mathcal{L}_{\text{NCE}}(\mathbf{S}_{\text{late}}(\mathcal{B}_{\text{txt}}), \tau) + w_3 \mathcal{L}_D(\mathcal{B}_{\text{txt}}) \\ & + w_4 \mathcal{L}_{\text{NCE}}(\mathbf{S}_{\text{dense}}(\mathcal{B}_{\text{multi}}), \tau) \\ & + w_5 \mathcal{L}_{\text{NCE}}(\mathbf{S}_{\text{late}}(\mathcal{B}_{\text{multi}}), \tau) + w_6 \mathcal{L}_D(\mathcal{B}_{\text{multi}}) \end{aligned}$$

Contrastive Loss

$$\mathcal{L}_{\text{NCE}}(B) := - \sum_{(x_i, y_i) \in B} \ln \frac{e^{s(x_i, y_i)/\tau}}{\sum_{i'=1}^k e^{s(x_i, y_{i'})/\tau}}$$

Distillation (Late $\square$ Dense)

$$\begin{aligned} \mathcal{L}_D(B, \tau) := & D_{\text{KL}}(\mathbf{S}'_{\text{dense}}(B) \parallel \mathbf{S}'_{\text{late}}(B)) \\ \text{where } \mathbf{S}'_{i,j} = & \text{softmax}(\mathbf{S}, \tau, i, j) \end{aligned}$$

+

Task-specific fine-tuning

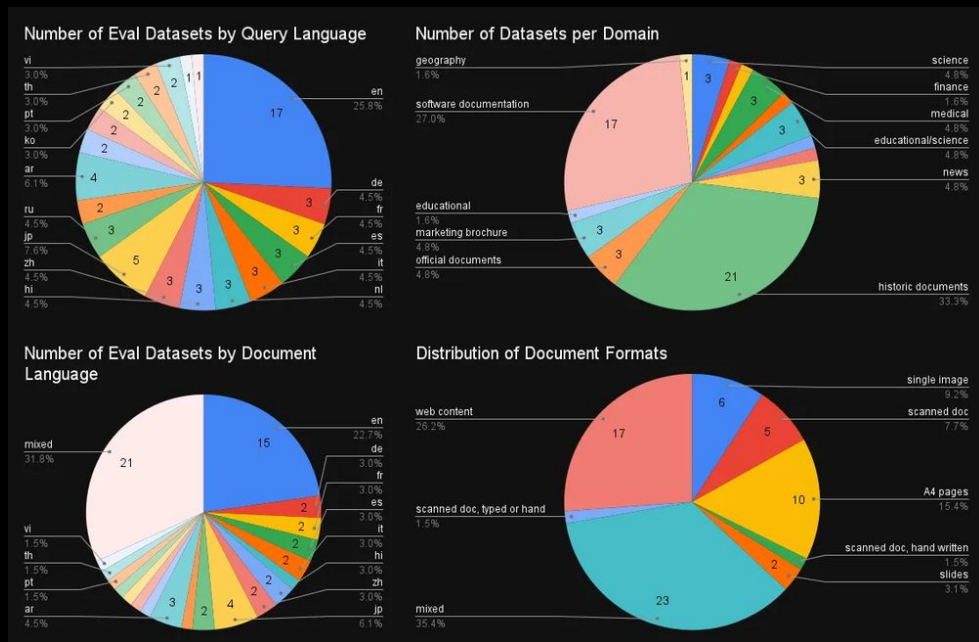
Benchmarking: Jina VDR

What do existing visual document retrieval benchmarks miss?

- Not multilingual
- Only one query type: Questions
- Mostly A4 PDF pages

New benchmark

- 95 tasks
- 20 languages
- Visual QA + keyword queries
- Diverse document types (charts, maps, docs, Markdown, tables)
- Many domains



Evaluation Results

Model	J-VDR	ViDoRe	CLIPB	MMTEB	MTEB-en	COIR	LEMB	STS-m	STS-en
jina-embeddings-v4 (dense)	73.98	84.11	84.11	66.49	55.97	71.59	67.11	72.7	85.89
jina-embeddings-v4 (late)	80.55	90.17	NA	NA	NA	NA	NA	NA	NA
text-embedding-3-large	--	--	--	59.27	57.98	62.36	52.42	70.17	81.44
bge-m3	--	--	--	55.36	NA	NA	58.73	NA	NA
multilingual-e5-large-instruct	--	--	--	57.12	53.47	NA	41.76	NA	NA
jina-embeddings-v3	47.82	26.02	--	58.58	54.33	55.07	55.66	75.77	85.82
voyage-3	--	--	--	66.13	53.46	67.23	74.06	68.33	78.59
gemini-embedding-001	--	--	--	67.71	64.35	73.11	NA	78.35	85.29
jina-embeddings-v2-code	--	--	--	NA	NA	52.24	NA	NA	NA
voyage-code	--	--	--	NA	NA	77.33	NA	NA	NA
nllb-clip-large-siglip	NA	NA	83.19	NA	NA	NA	NA	NA	NA
jina-clip-v2	40.52	53.61	81.12	NA	NA	NA	NA	NA	NA
colpali-v1.2 (late)	63.8	83.9	NA	NA	NA	NA	NA	NA	NA
dse-qwen2-2b-mrl-v1	67.25	85.8	NA	NA	NA	NA	NA	NA	NA
voyage-multimodal-v3	NA	84.24	NA	NA	NA	NA	NA	NA	NA

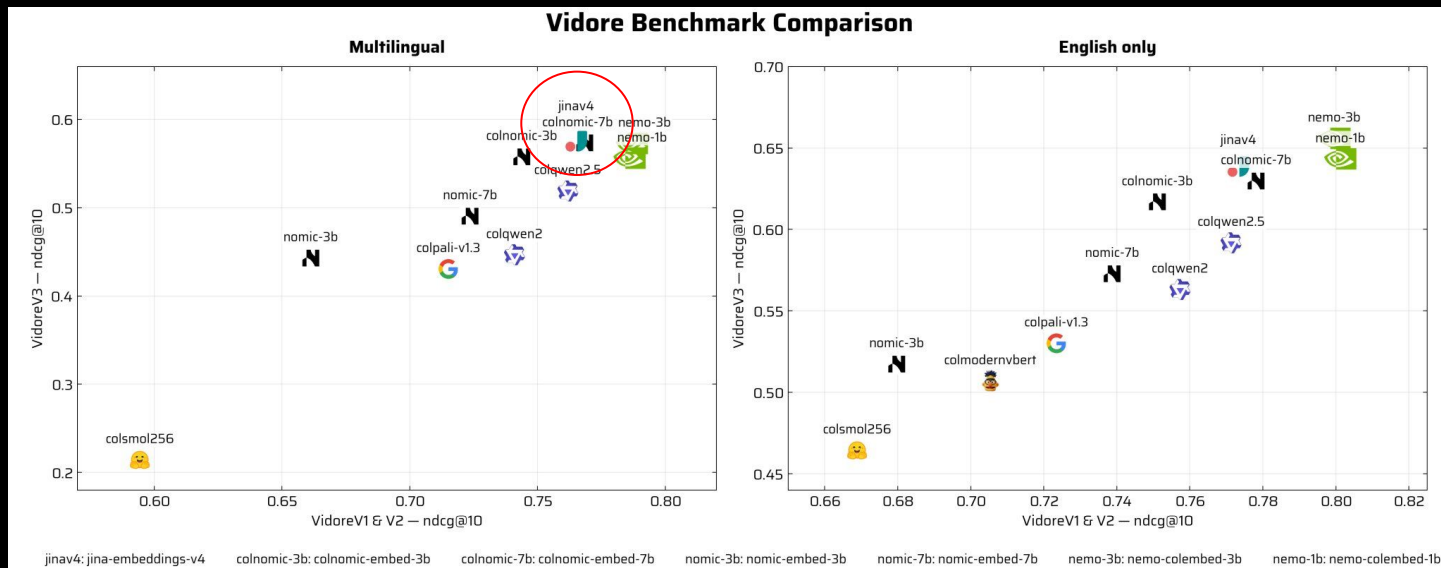
**Visual
Document
Retrieval**

**Picture
Retrieval**

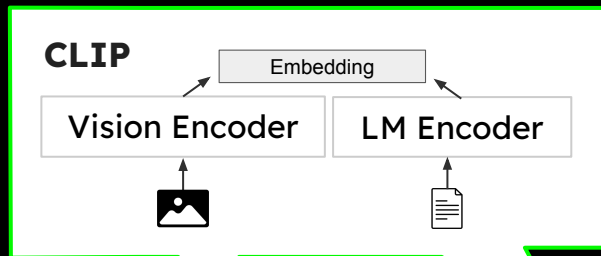
**Text
Retrieval**

**Semantic
Textual
Similarity**

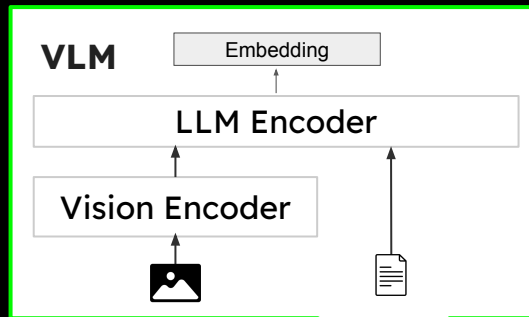
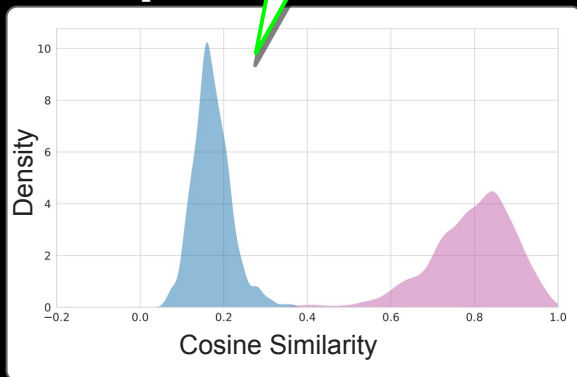
New Visual Document Retrieval Benchmark Vidore V3



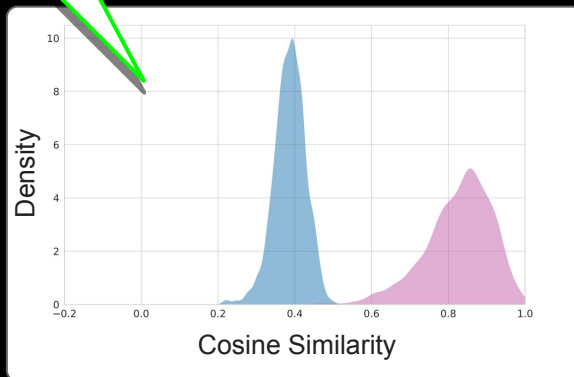
VLM Architecture Prevents Modality Gap



OpenAI CLIP



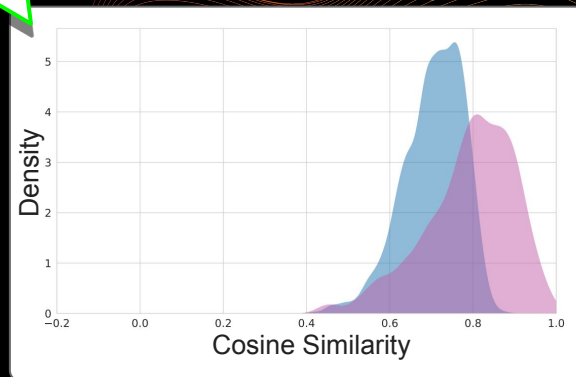
Jina CLIP



VLM Architecture uses a joined Encoder!

Text-to-Text
Text-to-Image

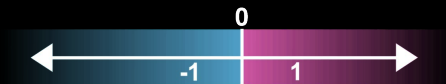
Jina-Embeddings V4



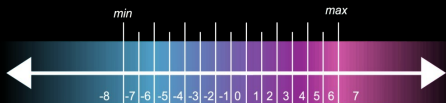
Vector Quantization/Quantization-Aware Training

Quantization Types

Binary: 0 and 1 represent -1 and 1



4- and 8-Bit: Each bit code covers a range



Quantization-aware training (QAT)

- Unlike post-training quantization (PTQ), continues training with quantization on same data
- Uses skip-through estimator to pass quantization step so gradients aren't lost

NanoBeIR retrieval task results

Quantization Type	AVG nDCG@10	Compared to Baseline
Baseline	60.1	n/a
PTQ Binary	58.33	-1.78
PTQ Binary Docs Only	59.08	-1.02
QAT Binary	59.22	-0.89
QAT Binary Docs Only	60.81	+0.70
QAT Trinary	59.49	-0.62
QAT 4-bits	61.73	+1.62
QAT 8-bits	61.67	+1.56
QAT 8-bits Min/Max	61.29	+1.19



<https://huggingface.co/jinaai/jina-embeddings-v4>

Jina VDR



<https://jina.ai/news/jinavr-news-visual-document-retrieval-benchmark-with-95-tasks-in-20-languages/>

Jina AI

 [oss.jina.ai](https://github.com/oss.jina.ai)

 discord.jina.ai



X @JinaAI_

Quantization-Aware Training

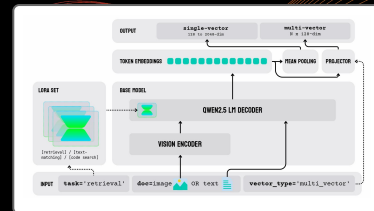


<https://jina.ai/news/quantization-aware-training-of-jina-embeddings-v4/>

Technical Report



<https://arxiv.org/abs/2506.18902>



<https://www.linkedin.com/in/ssturua/>